$$-\left(\frac{5}{4}\right)\ln|x|+\left(\frac{7}{2}\right)\frac{x^{-1}}{(-1)}+\left(\frac{5}{4}\right)\ln|x+2|+C=\int\left(\frac{\frac{-5}{4}}{x}+\frac{\frac{7}{2}}{x^2}+\frac{\frac{5}{4}}{x+2}\right)dx=$$

$$\int\left(-\left(\frac{5}{4}\right)\frac{1}{x}+\left(\frac{7}{2}\right)\frac{1}{x^2}+\left(\frac{5}{4}\right)\frac{1}{x+2}\right)dx=-\left(\frac{5}{4}\right)\ln|x|+\left(-\frac{7}{2x}\right)+\left(\frac{5}{4}\right)\ln|x+2|+C$$

Example 12.41: Find solution of the integral when there is a non-linear factor.

Integrate $\int\frac{x^5+1}{x^3(x+2)}dx$.

Ans: $\int\frac{x^5+1}{x^3(x+2)}dx=\int\frac{x^5+1}{x^4+2x^3}dx=\int(x-2+\frac{4x^3+1}{x^4+2x^3})dx=$

$$\int\left(x-2+\frac{A}{x}+\frac{B}{x^3}+\frac{C}{x^3}+\frac{D}{x+2}\right)dx$$

or, $Ax^2(x+2)+B(x+2)+C(x+2)+Dx^3=4x^3+1$

Let $x=0$ so that $A(0)+B(0)+C(2)+D(0)=1\to C=\frac{1}{2}$

let $x=-2$ so that $A(0)+B(0)+C(0)+D(-8)=-31\to D=\frac{31}{8}$;

let $x=1$ so that $A(3)+B(3)+C(3)+D=3A+3B+\frac{3}{2}+\frac{31}{8}=5\to A+B=-\frac{1}{8}$;

let $x=-1$ so that $A(1)+B(1)+C(1)+D(-1)=A-B+\frac{1}{2}-\frac{31}{8}\to A-B=\frac{3}{8}$;

it follows that $A=\frac{1}{8}$ and $B=-\frac{1}{4}$

Now $\int\left(x-2+\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{D}{x+2}\right)dx=\int\left(x-2+\frac{\frac{1}{8}}{x}+\frac{\frac{1}{4}}{x^2}+\frac{\frac{1}{2}}{x^3}+\frac{\frac{31}{8}}{x+2}\right)dx$

$$=\frac{x^2}{2}-2x+\frac{1}{8}\ln|x|-\left(\frac{1}{4}\right)\frac{x^{-1}}{(-1)}+\left(\frac{1}{2}\right)\frac{x^{-2}}{-2}+\frac{31}{8}\ln|x+2|+C=$$

$$\int\left(x-2+\frac{\frac{1}{8}}{x}+\frac{\frac{-1}{4}}{x^2}+\frac{\frac{1}{2}}{x^3}+\frac{\frac{31}{8}}{x+2}\right)dx=\frac{x^2}{2}-2x+\frac{1}{8}\ln|x|+\frac{1}{4x}-\frac{1}{4x^2}+\frac{31}{8}\ln|x+2|+C.$$

Example 12.42: Consider an improper rational function to evaluate the definite integral. We apply the technique of partial fractions first and then apply the limit values as has been done in cases with definite integrals.

$$\int_1^2\frac{x^3+4x^2-3x+3}{x^2-3x}dx$$

Ans: This rational function is improper because its numerator has a degree that is higher than its denominator. So, take $\frac{x^3+4x^2-3x+3}{x^2-3x}=(x-1)+\frac{-6x+3}{-x^2-3x}$.

Now apply partial function decomposition only on the remainder,

$$\frac{-6x+3}{-x^2-3x} = \frac{-6x+3}{-x(x-3)} = \frac{A}{x} + \frac{B}{x-3}$$

For solving the integration, the basic equation becomes $-6x+3 = A(x-3)+Bx$

For x = 0,

3 = –3A +B(0)

–1 = A

For $x = 3, \quad -18+3 = 0+3B$

or, $B = -5$

Thus, the integral becomes

$$\int_1^2 \frac{x^3+4x^2-3x+3}{x^2-3x}dx$$

$$= \int_1^2 \left[(x-1) + \frac{-6x+3}{-x^2-3x}\right]dx = \int_1^2 \left[x-1-\frac{1}{x}-\frac{5}{x-3}\right]dx$$

Integrating and substituting the limits,

$$= \left[\frac{x^2}{2} - x - \ln|x| - 5\ln|x-3|\right]_1^2$$

$$= \left(\frac{4}{2} - 2 - \ln|2| - 5\ln|1|\right) - \left(\frac{1}{2} - 1 - \ln|1| - 5\ln|2|\right)$$

$$= 4\ln|2| + \frac{1}{2}.$$

We take up integration of a trigonometric function to solve it with partial fractions.

Example:12.43

$$\int \sin^5 x dx = \int (\sin x)(\sin^2 x)^2 dx = \int (\sin x)(\sin^2 x)^2 dx = \int (\sin x)(1-\cos^2 x)^2 dx$$

Now by using substitution we can evaluate the integral by letting $u = \cos x$, then du = - sin x dx and

$$\int \sin^5 x dx = \int (\sin x)(\sin^2 x)^2 dx = \int (\sin x)(\sin^2 x)^2 dx$$

$$= \int (\sin x)(1-\cos^2 x)^2 dx$$

Using method of substitution

$$= \int -(1-u^2)^2 du = \int (-u^4 + 2u^2 - 1) du = \frac{-u^5}{5} + 2\frac{u^3}{3} - u + C$$

or, $\dfrac{-\cos^5 x}{5} + \dfrac{2\cos^3 x}{3} - \cos x + C$

Check Your Progress 7

1) Evaluate $\int \frac{3x+11}{x^2-x-6}dx$

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2) Find $\int \frac{2x+7}{x^2+4x+3}dx$

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3) Use partial fractions to evaluate: $\int \frac{6x+7}{(x+2)^2}dx$

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12.13 LET US SUM UP

We mainly discussed various techniques of integration in this unit. The concepts of indefinite and definite integrals were explained. Whereas the indefinite integral is a function and an anti-derivative, the definite integral is a number and can be used to represent the area under the curve. The processes of finding integrals of various functions through substitution, integration by part, trigonometric substitution and partial fractions are dealt with. Evaluation of improper integrals with infinite limits was also presented. Through out the unit, a lot of examples were shared in order to concretise the concept which helps in computing various integration problems.

12.14 KEYWORDS

Boundary Condition : A particular anti-derivative that satisfies certain initial condition.

Closed Interval : A closed interval is an interval that includes all its limit points. If the endpoints of the interval are finite numbers a and b, then the interval $x : a \le x \le b$ is denoted [a, b]. If one of the endpoints is $\pm\infty$, then the interval still contains all its limit points (although not all of its *endpoints*), so a, ∞ and $(-\infty, b)$ are also closed intervals, as is the interval $(-\infty, \infty)$.

Definite Integral : The definite integral of the function f(x) over the interval (a, b) is expressed symbolically as $\int_a^b f(x)dx$, read as "integral of *f* with respect to x from a to b. The smaller number a is termed the lower limit and b, the upper limit of integration. Geometrically, this definite integral denotes the area under the curve representing f(x) between the points x = 0 and x = b. Note that the definite integral $\int_a^b f(x)$ is a number.

Improper Integral : It is a special type of definite integral. When one of the limits of integration is $+\infty$ or $-\infty$ a definite integral is called an improper integral. Such integrals are evaluated using the concept of limits.

Indefinite Integral : The indefinite integral is basically reverse differentiation. To differentiate means to find the rate of change (derivative) is given.

Integration by Parts : A special method of integration that is often useful when two functions are multiplied together.

Partial Fractions : Each of two or more fractions into which a more complex fraction can be decomposed as a sum.

Rational Substitution : Integrand can be changed to a rational function by using substitutions.

Riemann Sum : A method for approximating the total area underneath a curve on a graph.

Substitution Rule : If g(x) is a differentiable function whose range is the interval I and f is continuous on I, then $\int f(g(x))g'(x)dx = \int f(u)du$ where u = g(x) and du = g'(x) dx.

12.15 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) We wish to perform the opposite process to differentiation and call this as "anti-differentiation". Thus, indefinite integral is a function whose derivative is a given function.
2) Using the rule of integration of a constant, we write: $\int 4dx = 4x + C$.
3) $\frac{1}{3}x^3 + 3x + C$
4) $-3e^{-3x} + C$
5) $\text{Ln}(x+2) + C$

Check Your Progress 2

1) $\frac{d}{dx}\sin ax = a\cos ax.$ We can say that $\int \cos ax dx = \frac{1}{a}\sin ax + C$ where a is an arbitrary constant and C is the constant of integration.
2) Always try for substitution to get the integration, if you cannot, go to integration by parts.
3) Hint: It simplifies the evaluation (derive the integral by taking u = ln x).

Check Your Progress 3

1) An integral expressed as the difference between the values of the integral at specified upper and lower limits of the independent variable.

2) Let a closed interval [a, b] be partitioned by points $a < x_1 < x_2 < ... < x_{n-1} < b$, where the lengths of the resulting intervals between the points are denoted $\Delta x_1, \Delta x_2, ..., \Delta x_n$. Let x_k^* be an arbitrary point in the kth subinterval. Then the quality $\sum_{i=1}^{n} f(x_k^*)\Delta x_k$ is called a Riemann sum for a given function f(x) and partition, and the value Δx_k is called the much size of the partition.

3) Answer yourself after reading the properties of definite integral.

Check Your Progress 4

1) It links the concept of a derivative to that of an integral. As a result, we can use our knowledge of derivatives to find the area under the curve, which is often quicker and simpler than using the definition of the integral.

2) They are opposite. If you take the derivative of a function, then integrate the result, you will end up with the function again. Definition-wise, the derivative of a function is the slope of a function, while the integral is the area between the function and the x-axis.

3) The definite integral of f(x) is a number and represents the area under the curve f(x) from x = a to x = b. Whereas the indefinite integral is a function f(x) and is an anti-derivative. In other words, it is a function F(x) whose derivative is f(x).

Check Your Progress 5

1) Integrating with the chain rule is the opposite of taking the derivative with the chain rule. For the derivative of the chain rule, $\frac{dy}{dx} a(g(x))^n = na(g(x))^{n-1} g'(x)$.

   So $\int na(g(x))^{n-1} g'(x) = a(g(x))^n$.

2) New limits are a = $2(-1)^2 + 3 = 5$ and b = $2(2)^2 + 3 = 11$

3) Put $u = 2x^2 + 3$ to get $du = 4xdx$

4) Now

$$\int_{-1}^{2} 4x(2x^2+3)^2 dx = \int_{-1}^{2} 4xu^2 = \frac{du}{4x} = \int_{-1}^{2}\left(4xu^2 \frac{du}{4x}\right) = \int_{5}^{11} u^2 du = \frac{1}{3}(u^3)|_5^{11}$$

$$= \frac{1}{3}(11^3 - 5^3) = \frac{1206}{3} = 402.$$

Check Your Progress 6

1) An improper integral is a definite integral that has either or both limits infinite or an integrand that approaches infinity at one or more points in the range of integration.

2) $\int_{-2}^{\infty} \sin x dx = \int_{b}^{0} \sin x dx = \lim_{b\to\infty}(-\cos x)|_{-2}^{b} = \lim_{b\to\infty}(\cos 2 - \cos b)|_{-2}^{b}$ As the limit does not exist, the integral is divergent.

3) $\int_{0}^{3} \frac{1}{(\sqrt{3-x})} dx = \lim_{3^-}\left(-2\sqrt{3-x}\right)|_{0}^{b} = \lim_{3^-}(2\sqrt{3}) - 2(\sqrt{3}-b) = 2\sqrt{3}$. The limit exists and is finite and so the integral converges and the integral's value is $2\sqrt{3}$.

Check Your Progress 7

1) $\int \frac{3x+11}{(x-3)(x+2)} dx = \int\left(\frac{A}{(x-3)} + \frac{B}{(x+2)}\right) dx$. Now work with $\frac{3x+11}{(x-3)(x+2)} =$

$\left(\frac{A}{(x-3)} + \frac{B}{(x+2)}\right) = \left(\frac{A(x+2)+B(x-3)}{(x-3)(x+2)}\right)$ to get $3x+11 = A(x+2)+B(x-3)$

Let x = -2 so that $5 = A(0)+B(-5) \to B = -1$ and $x = 3$ so that $20 = A(5)+B(0) \to A = 4$.

Putting the value of A and B in the integration we have

$$\int \frac{3x+11}{(x-3)(x+2)} dx$$

$$= \int\left(\frac{4}{(x-3)} + \frac{-1}{(x+2)}\right) dx$$

$$= \int\left(\frac{4}{(x-3)}\right) dx + \left(\frac{-1}{(x+2)}\right) dx$$

$$4\ln|x-3| - \ln|x+2| + C$$

2) Set up partial fractions: $\frac{2x+7}{x^2+4x+3} = \frac{A}{1+x} + \frac{B}{x+3}$ so that $2x+7 = A(x+3)+B(x+1)$.

Solve for $x=-1$ and $x=-3$ to get $A = \frac{5}{2}$ and $B = \frac{-1}{2}$

Ans.: $\frac{5}{2}\ln|x+1| - \frac{1}{2}\ln[x+3] + C$.

3) Decompose partial fraction: $6x+7 = A(x+2)+B$.

Now $\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$. If $x=-2$, then $B=-5$; if $x = 0$, then $A = 6$.

So, $\int \frac{6x+7}{(x+2)^2} dx = \int \frac{6}{x+2} dx - \int \frac{5}{(x+2)^2} dx$. Put $u = x+2$ and solve to get the answer as $6\ln|x+2| + \frac{5}{x+2} + C$.

12.16 EXERCISES

Q1. Evaluate the following integral: $\int \sin^6 x \cos^3 x dx$

Ans. Hint: $\int \sin^6 x \cos^3 x dx = \int \sin^6 x \cos^2 x \cos x dx = \int \sin^6 x(1-\sin^2 x)\cos x dx$.

Put sin x = u to get $\int u^6 x(1-u^2) du$. You can integrate the function now.

Q2. Evaluate the following integral:

$$\int \frac{1}{2}((1-\cos(2x))\frac{1}{2}(1+\cos(2x)))dx$$

Ans. Hint: write $\int \sin^2 x \cos^2 x dx = \int \frac{1}{2}((1-\cos(2x))\left(\frac{1}{2}\right)((1+\cos(2x))dx =$

$\frac{1}{4}\int 1-\cos^2(2x)\,dx = \frac{1}{4}\int 1-\frac{1}{2}(1+\cos(4x))dx$ to get the answer $\frac{1}{8}x - \frac{1}{32}\sin(4x) + C$.

Q3. Evaluate the following integral:

Ans.: $\int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x}dx = \int \frac{(\sec^2 x + \tan x \sec x)}{\sec x + \tan x}dx$.

Put $\sec x + \tan x = u$ for integration which yields $\ln|\sec x + \tan x| + C$.

Q4. Evaluate the integral: $\int xe^{6x}dx$

Ans. See that choosing u = x will help as on differentiating x will drop out.

Since u = x, du = dx.

Put $dv = e^{6x}dx$ so that $v = \int e^{6x}dx = \frac{1}{6}e^{6x}$

The integral is then $\int e^{6x}dx = \frac{x}{6}e^x dx - \int \frac{1}{6}e^{6x}dx = \frac{x}{6}e^x - \frac{1}{36}e^{6x} + C$

Q5. Evaluate the following integral $\int x\sqrt{x+1}\,dx$

a) Using Integration by Parts.

b) Using substitution.

Ans: (a) Evaluate using Integration by parts.

We select the choices for u and dv to handle the problem at hand.

Take u = x and $dv = \sqrt{x+1}\,dx$ so that du = dx and $v = \frac{2}{3}(x+1)^{\frac{3}{2}}$

Then the integral is $\int x\sqrt{x+1}\,dx = \frac{2}{3}(x+1)^{\frac{3}{2}} - \frac{2}{3}\int (x+1)^{\frac{3}{2}}(x+1)^{\frac{3}{2}}dx$

$= \frac{2}{3}(x+1)^{\frac{3}{2}} - \frac{4}{15}(x+1)^{\frac{5}{2}} + c$

b) Evaluate using method of substitution

Take u = x + 1 so that x = u – 1 and du = dx.

Note that you need to use the substitution twice, once for the quantity under the square root and once for the x in front of the square root. The integral is then,

$$\int x\sqrt{x+1}\,dx = \int (u-1)\sqrt{u}\,du = \int u^{\frac{3}{2}} - u^{\frac{1}{2}}\,du = \frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}} + C.$$

$$= \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + C$$

Q6. Integrate $\int \frac{\ln x}{x^5}dx$

Ans. Hints: Put $u = \ln x$ and $dv = \frac{1}{x^5}dx$. Final answer is $\frac{\ln x}{x^4} = \frac{1}{16x^4} + C$

Q7. Integrate $\int x^2 e^{3x} dx$

Ans. Hints: Put $u = x^2$ and $dv = e^{3x}dx$. Final answer is $\frac{1}{3}x^2e^{3x} - \frac{2}{9}x^2e^{3x} + \frac{2}{27}e^{3x} + C$

Integrate $\int \frac{\ln x}{x^5}dx$

Ans. Hints: Put $u = \ln x$ and $dv = \frac{1}{x^5}dx$. Find answer is $\frac{\ln x}{x^4} - \frac{1}{16x^4} + C$.

Q8. Integrate $\int x^2 e^{3x} dx$

Ans. Hints: Put $u = x^2$ and $dv = e^{3x}dx$. Find answer is $\frac{1}{3}x^2e^{3x} - \frac{2}{9}x^2e^{3x} + \frac{2}{27}e^{3x} + C$.

Q9. Determine if the following integral is convergent or divergent. If it is convergent find its value.

$$\int_{2}^{3} \frac{1}{x^3}dx$$

Ans. This integrand is not continuous at x = 0.So,we'll split it at that point.

Thus, $\int_{-2}^{3} \frac{1}{x^3}dx = \int_{-2}^{0} \frac{1}{x^3}dx + \int_{0}^{3} \frac{1}{x^3}dx$

Now we need to look at each of these integrals and see if they are convergent.

$$\int_{-2}^{0} \frac{1}{x^3}dx = \lim_{b\to 0^-} \int_{-2}^{b} \frac{1}{x^3}dx$$

$$= \lim_{b\to 0^-}\left(-\frac{1}{2x^2}\right)\Bigg|_{-2}^{b} = \lim_{b\to 0^-}\left(-\frac{1}{2b^2} + \frac{1}{8}\right) = -\infty$$

One of the integrals is divergent which means the given integral is divergent. We do no need to examine the second integral.

Q10. Find solution of $\int \frac{2-x}{x^2+5x}dx$

Ans.: $\frac{2}{5}\ln|x| - \frac{7}{5}\ln|x+5| + C$

Q11. Find solution of $\int \frac{2-x}{x^2+5x}dx$

Ans.: $\int \frac{2-x}{x^2+5x}dx = \int\left(1+\frac{15}{x^2-16}\right)dx$ will give $A(x-4)+B(x+4)=15;$

Putting $x=-4$ and $x=4$ you get $A=-15/8$ and $B=15/8$ leading to the solution of

$x+\frac{15}{8}(\ln|x-4|-\ln|x+4|)+C$.

NOTES